

UNIVERSITY NAME

*Analysis IV — Final Exam**Year: 2024***Exercise 1:**

Determine the nature (convergence/divergence) of the following numerical series of general term:

$$\begin{aligned} a) \ u_n &= \frac{n^n}{(2n)!} \\ b) \ u_n &= \frac{n + \cos n}{e^n + \sin n} \\ c) \ u_n &= \sqrt{1 + \frac{(-1)^n}{\sqrt{n}}} - 1 \end{aligned}$$

Answer Area

Exercise 2:

Consider the series of functions $\sum b_n$ defined by:

$$f_n(x) = \frac{1}{n + n^2 x}$$

We denote:

$$f(x) = \sum_{n=1}^{\infty} f_n(x)$$

1. Determine the domain of definition of f .
2. Show that f is continuous on $\mathbb{R}_+^*$ and compute:

$$\lim_{x \rightarrow \infty} f(x); \lim_{x \rightarrow 0} f(x)$$

deduction.

3. Show that f is of class C^1 on $\mathbb{R}_+^*$ and give its derivative.
4. Sketch the variation table of f on $\mathbb{R}_+^*$ and draw its graph.

5. (a) Show the existence and compute the integral:

$$\int_1^\infty \frac{1}{y + y^2 x} dy$$

- (b) Show that:

$$\log \frac{x+1}{x} \leq f(x) \leq \log \left(\frac{1+2x}{2x} \right) + \frac{1}{1+x}$$

Deduce an equivalent of $f(x)$ at 0.

Answer Area

Exercise 3:

Let $f : x \in (-1, 1) \longrightarrow \frac{\arcsin x}{\sqrt{1-x^2}}$.

1. Justify that f can be expanded as a power series.
2. Show that f satisfies a first-order differential equation.
3. Deduce the power series expansion of f .
4. Deduce that for any integer n :

$$\sum_{k=0}^n \frac{1}{2k+1} \binom{2k}{k} \binom{2(n-k)}{n-k} = \frac{2^{4n}(n!)^2}{(2n+1)!}$$

It is recalled that:

$$\frac{1}{\sqrt{1-x^2}} = \sum_{n=0}^{\infty} \frac{(2n)!}{2^{2n}(n!)^2} x^{2n}$$

Answer Area